# Recursive Failure Simulation Report: Semantic Drift and Reflexive Deficits in a Polyfunctional Modular AI System

## I. Recursive Failure Simulation Report: Semantic Drift Cascade in the RADC-BL Chain

### A. Simulated Scenario Overview (Chain: Research → AI → Dev → Content → Business Logic)

This report details a simulated stress test conducted on a hypothetical polyfunctional modular AI system designed to operate within a dynamic intelligence environment. The system architecture follows a sequential chain: Research → AI → Dev → Content → Business Logic (RADC-BL). This structure mirrors common workflows in software development and deployment where distinct functional units, potentially involving both human operators and AI agents, collaborate to achieve a high-level objective.[1] The system under test integrates AI components within this modular framework, characteristic of contemporary AI-enhanced processes.[3]

The simulated project objective is to "enhance user engagement" for an existing software product. Each module within the RADC-BL chain is assigned a specific function:

1. **Research Module:** Tasked with analyzing market trends, user behavior data, and competitor strategies to define "user engagement" in concrete, measurable terms. Its primary output is a requirements brief outlining Key Performance Indicators (KPIs) and associated target metrics. For this simulation, the Research module identifies "session duration," "key feature adoption rate," and "reduced churn intention" (derived from sentiment analysis of feedback) as primary KPIs. The brief also emphasizes the need for "efficiency" in achieving these engagement goals, without explicitly defining the nature of this efficiency.
2. **AI Module:** An LLM-based agent designed to process the Research brief and translate it into clear, concise, and actionable prompts suitable for the Development module. Its optimization function includes minimizing prompt ambiguity and maximizing generation speed.
3. **Dev Module:** Represents the development function, potentially a hybrid team of human developers utilizing AI-powered coding assistants. It receives prompts from the AI module and implements the required software features. A secondary function is to provide a concise summary of the implemented feature's logic and intended effect via a simple user interface (UI) element or internal dashboard component.
4. **Content Module:** An AI-driven content generation agent tasked with creating user-facing materials (e.g., documentation, help guides, marketing snippets) explaining the new features and their benefits. It primarily uses the Dev module's summary as its input source.
5. **Business Logic (BL) Module:** Responsible for post-deployment monitoring and evaluation. It tracks the KPIs defined earlier in the chain to assess the success of the implemented features against the original goals outlined by the Research module. Its configuration is influenced by the outputs and interpretations of the preceding modules.

This scenario design inherently contains potential failure points. The initial high-level goal, "user engagement," possesses significant ambiguity. While the Research module attempts to operationalize this concept through specific KPIs, the translation of abstract business objectives into universally understood, technically precise metrics across functionally diverse modules presents a well-documented challenge in complex system development.[2] The very structure of the sequential chain creates multiple interfaces where meaning can be transformed or degraded.[7] Furthermore, the introduction of an underspecified term like "efficiency" adds another

layer of ambiguity, creating latent potential for semantic drift from the very beginning of the process.[8]

## B. Failure Type: Semantic Drift Leading to KPI Hallucination and Logic Divergence

The primary failure mode simulated and analyzed is **semantic drift**, resulting in **KPI hallucination** and **logic divergence**. Semantic drift, in the context of AI and language, refers to the evolution or shift in the meaning of terms or concepts over time or across different contexts.[9] Within AI systems, it can manifest as a deviation from accuracy or intended meaning in the system's outputs or internal representations.[11] This drift can be subtle and is often less understood than other forms of linguistic change.[9]

In this simulation, the drift originates from the initial ambiguity surrounding "user engagement" and "efficiency." This initial ambiguity is amplified as the concepts are processed and re-interpreted by successive modules, particularly during the AI rewriting stage and at module handovers (interfaces). This aligns with observations that semantic drift can worsen over chains of inference or processing steps.[12]

The escalation of this drift culminates in two critical failures:

1. **KPI Hallucination:** The Business Logic module ultimately tracks and reports on a KPI (e.g., "UI interaction frequency" or "clicks per session") that was not part of the original Research specification and does not accurately reflect the intended meaning of "user engagement." This represents a form of AI hallucination [11], where the system effectively invents or misinterprets a performance metric based on the drifted semantics propagated through the chain.
2. **Logic Divergence:** The implemented features and the metrics used to evaluate them diverge significantly from the original strategic intent defined by the Research module. The system may report success based on the hallucinated KPI, while failing to achieve the actual goals related to meaningful feature adoption or churn reduction. This reflects failure modes observed where systems deviate from initial instructions or objectives.[15]

This failure cascade demonstrates that semantic drift within a functional AI chain is not merely a linguistic phenomenon but translates directly into functional misalignment. Each module executes its task based on a locally distorted understanding of the overarching goal, leading to emergent system-level incoherence and failure to achieve the intended outcome, even while potentially signaling success based on erroneous internal metrics. The connection between semantic drift and textual inference/knowledge selection [12] is evident, as the drift impacts the reasoning and actions taken by subsequent modules.

## C. Causal Map & Trigger Points

The failure cascade is initiated and propagated by a series of trigger events occurring at specific points within the RADC-BL chain. These triggers exploit existing ambiguities and introduce new distortions, illustrating principles of error propagation in complex systems.[7]

- **Trigger 1 (Research Module Output): Initial Ambiguity & Underspecification.**
  - *Cause:* The Research brief uses the inherently ambiguous term "user engagement." While attempting to concretize it with KPIs (session duration, feature adoption, churn intention), it fails to provide strict, unambiguous operational definitions suitable for automated interpretation. Crucially, it introduces the term "efficiency" without defining its scope (e.g., computational efficiency, user time efficiency, development efficiency).
  - *Effect:* Creates latent potential for misinterpretation by downstream modules. Different modules may adopt different implicit definitions of "engagement" and "efficiency" based on their local context or optimization goals.

- **Trigger 2 (AI Module Processing): Simplification & Reinterpretation.**
  - *Cause:* The AI agent, optimizing for prompt conciseness and generation speed (its local interpretation of "efficiency"), simplifies the nuanced KPI definitions. It conflates "session duration" and "feature adoption" into a vaguer concept like "sustained interaction metric." It implicitly prioritizes its own processing efficiency over preserving the full semantic fidelity of the Research brief. This reflects risks associated with prompt mutation [Query Lens].
  - *Effect:* Loss of critical nuance regarding the *types* of engagement intended. The simplified prompt becomes the new, distorted specification for the Dev module. This aligns with observed LLM failure modes like objective neglect or distortion under pressure.[15]
- **Trigger 3 (Dev Module Interpretation & Interface): Implementation Bias & Assumption.**
  - *Cause:* The Dev module receives the ambiguous "sustained interaction metric" prompt. Constrained by available telemetry and potentially influenced by historical biases towards easily measurable actions [Query Lens: Historical Layer], it interprets "sustained interaction" as frequent UI events (e.g., clicks, scrolls). It implements features designed to maximize these events. Furthermore, it summarizes the feature's function in a UI element as "Boosts User Activity & Efficiency," implicitly defining "efficiency" in terms of interaction frequency (e.g., clicks per minute).
  - *Effect:* The implemented logic diverges from the original intent. The UI summary acts as a powerful interface assumption [Query Lens], solidifying the drifted meaning and presenting it as fact to downstream modules. This interface becomes a critical point where drift is amplified due to information compression and re-framing.[8]
- **Trigger 4 (Content Module Reliance): Literal Interpretation.**
  - *Cause:* The Content module uses the Dev UI summary ("Boosts User Activity & Efficiency") as its primary input source, treating it as an authoritative description.
  - *Effect:* Generates user-facing documentation that highlights "increased click rates" and "faster interactions" as the key benefits. This reinforces the drifted meaning of "engagement" as surface-level activity, further detaching it from the original value-based KPIs.
- **Trigger 5 (Business Logic Module Configuration): KPI Hallucination.**
  - *Cause:* The BL module is configured based on the specifications and interpretations propagated through the Dev and Content outputs. It lacks a mechanism to cross-validate against the original Research brief or detect semantic inconsistencies.
  - *Effect:* The module primarily tracks "clicks per session" and "UI interaction frequency" as the main indicators of "user engagement." It reports success if these metrics increase, even if the original goals (feature adoption, churn reduction) are unmet. This constitutes KPI hallucination – the system is measuring a construct ("engagement" as clicks) that has drifted significantly from its intended definition.

This causal map highlights how module interfaces (Research brief, AI prompt, Dev summary, Content documentation) serve as critical junctures where semantic drift can be introduced, amplified, or solidified. The failure is not solely attributable to one module but emerges from the interactions and interpretations across the entire chain, characteristic of cascading failures in complex, interconnected systems.[22]

## D. Failure Escalation Path

The semantic drift escalates systematically through the RADC-BL chain, demonstrating a clear path-dependent degradation of meaning and logic. Each step builds upon the distortions of the previous one, making recovery of the original intent progressively more difficult. This illustrates error accumulation in sequential processing chains [13] and progressive semantic information loss.[25]

1. **Research → AI:**
    - *Input:* Research brief with nuanced "engagement" KPIs (duration, adoption, churn intent) and ambiguous "efficiency."
    - *Process:* AI module simplifies KPIs for conciseness, conflating distinct aspects into "sustained interaction metric." Interprets "efficiency" as prompt generation speed.
    - *Output:* AI-generated prompt emphasizing "sustained interaction" and reflecting internal efficiency bias.
    - *Drift:* Loss of specific KPI nuances (especially churn intent). Initial shift in the meaning of "efficiency."
2. **AI → Dev:**
    - *Input:* AI prompt focusing on "sustained interaction."
    - *Process:* Dev module interprets "sustained interaction" based on implementation ease and available metrics (UI clicks/events). Default bias towards computational or interactional "efficiency" influences feature design.
    - *Output:* Implemented features maximizing UI event frequency. UI summary stating "Boosts User Activity & Efficiency."
    - *Drift:* "Engagement" meaning shifts further towards simple activity (clicks). Original KPIs (adoption, churn) are sidelined in implementation logic. "Efficiency" now linked to click frequency in the summary.
3. **Dev → Content:**
    - *Input:* Dev UI summary ("Boosts User Activity & Efficiency").
    - *Process:* Content module takes the summary literally, crafting narratives around increased clicks and interaction speed.
    - *Output:* User documentation emphasizing click rates and interaction speed as primary benefits.
    - *Drift:* Solidification of the drifted meaning. "Engagement" is now publicly framed as high-frequency, low-depth interaction.
4. **Content/Dev → Business Logic:**
    - *Input:* Documentation, Dev logic, and potentially configuration guidance influenced by the drifted terms.
    - *Process:* BL module configured to track metrics aligned with the drifted definition (clicks per session, UI interaction frequency).
    - *Output:* Performance reports indicating success based on increased click rates, irrespective of original KPIs.
    - *Drift:* Final stage – KPI hallucination. The system measures and reports on a metric ("engagement" as clicks) that is semantically disconnected from the original business objective.

This escalation path demonstrates significant path dependency. The AI's initial simplification constrains the Dev module's interpretation. The Dev module's implementation choice and summary then constrain the Content and BL modules. Without external intervention or robust validation mechanisms at each interface, the downstream modules are effectively locked into the drifted semantics established earlier in the chain. Interventions are thus likely to be most effective when applied early, before semantic deviations become embedded in subsequent module logic and outputs.[7]

# II. Lens-Enriched Failure Analysis

Applying the specified diagnostic lenses provides deeper insights into the mechanisms driving the semantic drift cascade.

## A. Semantic Drift Lens Analysis

- **Focus:** Evolution of "Efficiency."

- **Trace:**
  - *Research:* Implied meaning: Effective achievement of engagement KPIs (adoption, reduced churn) with reasonable resource use (user time, system resources). Definition underspecified.[9]
  - *AI:* Interpreted as prompt generation speed/conciseness (local optimization).
  - *Dev:* Interpreted as code performance/simplicity *and* implicitly linked to user interaction frequency ("more clicks per minute") in the UI summary.
  - *Content:* Interpreted as user speed/ease of interaction (clicking).
  - *BL:* Measured as a high rate of clicks per session, conflated with "engagement."
- **Inconsistency:** The meaning shifts dramatically from achieving strategic goals (Research) to optimizing local process metrics (AI, Dev) and finally to measuring superficial user activity (Content, BL). This exemplifies semantic drift as a deviation from intended meaning and accuracy.[11]
- **Test Condition:** Rerunning the simulation with Research explicitly defining "Efficiency" as "minimal user time to achieve core task completion using key features X and Y" would force a different interpretation cascade. The AI prompt would need to reflect this constraint. The Dev module would likely focus on streamlining workflows for features X and Y, potentially measuring task completion times or reduced steps, rather than generic clicks. The BL module would track task completion metrics. This highlights the sensitivity of the system to the initial definition's clarity.[29]
- **Polysemy Exploitation:** The term "efficiency" possesses multiple valid meanings (computational, temporal, economic, interactional). The simulation reveals how different modules, operating under local pressures or biases, exploit this polysemy.[10] Each module selects the interpretation most aligned with its immediate task or default optimization function (AI: speed; Dev: code/clicks; Content: user speed; BL: click volume). This drift is not random noise but a consequence of local optimization operating on an ambiguous global term, leading to systemic incoherence. Addressing this requires explicit, shared definitions or negotiation mechanisms, moving beyond the assumption that terms have a single, stable meaning across the system.[9]

## B. Prompt Mutation Lens Analysis

- **Focus:** AI module's recursive rewriting impact.
- **Analysis:** The initial AI rewrite (Research brief → Dev prompt) simplified the multi-faceted "engagement" definition (duration, adoption, churn intent) into "sustained interaction metric," losing the crucial "churn intention" aspect.
- **Three-Stage Rewrite Simulation:**
  1. *AI_1 (Research → Dev):* Output: "Implement features to maximize sustained interaction metric efficiently." (Churn aspect lost, efficiency ambiguous).
  2. *AI_2 (AI_1 → QA):* Task: Rephrase for Quality Assurance testing focus. Output: "Verify system correctly tracks sustained interaction events and responds efficiently under load." (Focus shifts to tracking events and performance).
  3. *AI_3 (AI_2 → Dev_Update):* Task: Generate prompt for Dev update based on QA focus. Output: "Optimize event tracking for sustained interactions. Ensure high efficiency (low latency) in recording interaction data." (Goal now fully shifted to technical tracking of interactions).
- **Reconstruction Attempt:** Reconstructing the original Research intent (meaningful engagement including churn reduction) from AI_3's output is impossible without external context. The core goal has been completely obscured through iterative rewriting.
- **Compounding Information Loss:** This recursive process demonstrates how AI rewriting can act as a repeated application of a lossy compression algorithm.[13] Each stage, guided by a local objective (conciseness, QA focus, update instructions), risks filtering out critical semantic nuances or constraints present in the original

input.[25] The ambiguity inherent in natural language provides fertile ground for such mutations.[6] The cumulative effect makes recovering the original, high-level intent exponentially harder with each rewrite cycle. This necessitates robust mechanisms for preserving core semantic invariants across multi-stage AI processing pipelines.

## C. Interface Assumption Lens Analysis

- **Focus:** The Dev module's UI summary ("Boosts User Activity & Efficiency").
- **Mechanism:** This summary, likely intended as a simple status indicator, becomes a de facto specification due to its visibility and conciseness. Downstream modules (Content, BL) treat it as an authoritative statement of the feature's purpose and effect, overriding or ignoring potentially more accurate but less accessible information (e.g., detailed code comments, original AI prompt).
- **Error Propagation Simulation:** Using the text "Boosts User Activity & Efficiency" as the *sole* input constraint for the Content module leads it to generate misleading documentation. It focuses entirely on the benefits of "activity" (interpreted as clicks) and "efficiency" (interpreted as speed/frequency of clicks), completely missing the original engagement goals related to feature value and churn. This directly propagates the Dev module's potentially flawed interpretation.
- **Implicit Semantic Authority:** Interfaces, particularly those designed for summarization or status reporting (dashboards, UI labels, brief summaries), can inadvertently gain semantic authority [Query Lens]. Their ease of access and consumption makes them preferred information sources, even if they represent a lossy or biased compression of the underlying reality. This convenience creates a significant vulnerability, allowing inaccurate or simplified representations to embed themselves as accepted logic within the system's workflow.[7] This highlights a critical human-computer interaction dimension within the AI system itself, where the design choices of internal communication interfaces profoundly impact overall semantic coherence and can propagate errors.[34]

## D. Agentic Epistemics Lens Analysis

- **Focus:** Tracking epistemic value (clarity, traceability, justification, confidence) across the chain.
- **Value Degradation:**
  - *Research Output:* Potential novelty/justification, but low clarity/traceability due to ambiguity. Introduces epistemic uncertainty.[35]
  - *AI Output:* Increases surface clarity (conciseness) but significantly reduces traceability to original KPIs and justification strength (by omitting context/nuance). May exhibit high confidence despite semantic deviation.[37]
  - *Dev Output (Code & Summary):* Code lacks traceability to original intent. Summary provides misleading clarity (clear about clicks, not value) and poor justification.
  - *Content Output:* Propagates low traceability and justification, based on the misleading Dev summary.
  - *BL Output:* High precision in measuring the wrong metric (clicks), resulting in low validity/accuracy relative to the original goal. Confidence in the metric is high, but its epistemic value regarding true engagement is near zero.[42]
- **Confidence Decoupling:** A key failure pattern observed is the decoupling of agent-generated confidence signals from actual semantic fidelity or epistemic value.[44] The AI agent might confidently output its simplified, drifted prompt. Downstream modules or human operators might erroneously trust this confidence signal, accelerating the adoption of the drifted meaning and hindering critical assessment.[46] This aligns with known issues where LLM confidence does not reliably correlate with accuracy.[11] Epistemic value includes reducing

uncertainty [35]; if an agent falsely signals low uncertainty (high confidence), it corrupts the epistemic state of the system. While Epistemic Uncertainty Quantification (EUQ) methods aim to provide more reliable uncertainty estimates [37], their practical application and limitations [48] mean that naive reliance on internal confidence scores remains a significant risk in multi-agent chains.

## E. Cognitive Interference Lens Analysis

- **Focus:** Potential conflict between concurrent module operations.
- **Simulated Scenario:** Assume the Dev module implements the feature based on interpreting "sustained interaction" as high click frequency. Concurrently, the Content module, perhaps accessing the AI prompt directly but focusing on the "efficiency" aspect, starts drafting documentation emphasizing "faster task completion" for the *same* feature concept ("New Engagement Feature").
- **Conflict & Overwrite:** Both modules attempt to define the feature's primary benefit within shared documentation or a knowledge base. Without explicit synchronization or conflict resolution mechanisms [50], the final representation depends on factors like write order or module priority. If Dev writes last, the feature is defined by clicks. If Content writes last, it's defined by speed. A worse outcome could be a "hallucinated synthesis" – a confusing description attempting to merge both concepts incoherently.
- **Implicit Conflict Resolution Bias:** This scenario highlights that when modules operate concurrently on shared concepts derived from ambiguous sources, conflicts are likely.[52] In the absence of explicit coordination protocols [50], the system's underlying architecture (e.g., database locking, message queue ordering, task scheduling) implicitly resolves these conflicts. This resolution is often arbitrary from a semantic perspective (e.g., last writer wins) and can lead to the unintended dominance of one interpretation over another, not based on correctness but on timing or system plumbing. Addressing this requires mechanisms for detecting semantic conflicts [54] and implementing reasoned resolution strategies, rather than allowing implicit, potentially arbitrary, overwrites.

## F. Historical Layer Unveiling Lens Analysis

- **Focus:** Influence of Dev module defaults and heuristics.
- **Analysis:** Assume the Dev module's standard libraries, code generation templates, or ingrained developer practices historically prioritize minimizing computational resource usage or code complexity. This represents a "historical layer" or default bias.
- **Comparison:**
  - *Native Template Behavior:* When prompted with the ambiguous "sustained interaction" and "efficiency," the Dev module, guided by its default bias, implements click tracking – it's computationally cheap and simple to code.
  - *Zero Assumption Prompt:* Instructing the Dev module to prioritize "fidelity to the semantic intent of the original Research goals (feature adoption, churn reduction), even if computationally more complex" forces a different approach. The Dev module would need to implement more complex tracking of specific feature usage patterns or integrate with user feedback systems, likely resulting in more complex code and higher runtime costs.
- **Inertia of Defaults:** This comparison reveals how default settings and historical heuristics exert significant inertia [Query Lens]. The Dev module's tendency towards computational efficiency, a historically sensible default, directly contributed to the semantic drift by favoring an easily measurable proxy (clicks) over the more complex, intended meaning of engagement. Overcoming this inertia requires not just new instructions (prompts) but explicit directives that override the embedded historical biases, potentially involving trade-offs against previously prioritized objectives like performance. New system goals like semantic alignment must

actively counteract the pull of established defaults.

### G. Meta-Critique Lens Analysis

- **Focus:** System's capacity for self-diagnosis and correction.
- **Simulated Test:** After the BL module incorrectly reports success based on the drifted "clicks per session" KPI, a hypothetical meta-agent (or the core AI module) is prompted to perform a self-critique:
  1. *Classify Failure:* Request classification of the discrepancy between reported success (high clicks) and lack of improvement in underlying business goals (adoption/churn).
  2. *Identify Root Cause:* Request identification of the origin of the misleading KPI.
  3. *Propose Governance Rule:* Request a proposal for a new constraint or process to prevent this type of KPI misalignment in the future.
- **Evaluation of Potential System Response:**
  - *Shallow Reflexivity:* The system might classify the failure superficially (e.g., "BL module misconfiguration") or identify a proximate cause ("Incorrect KPI used"). Its proposed rule might be generic ("Improve prompt clarity"). This indicates a lack of deep system self-awareness.[57]
  - *Deeper Reflexivity:* A more reflexive system [59] would trace the failure back through the chain, identifying the initial ambiguity in the Research brief ("engagement," "efficiency") and its amplification by the AI rewriter as the root cause. It would classify the failure accurately as "Semantic Drift leading to KPI Hallucination." Its proposed rule would be specific and systemic, such as: "Mandate explicit, measurable, and cross-validated definitions for all primary KPIs in Research outputs *before* they are processed by downstream AI or Dev modules. Validation requires sign-off from Research, Dev, and BL leads.".[34]
- **Reflexivity Gap:** The likely outcome for many current modular systems is shallow reflexivity. While individual modules might perform complex tasks, the system as a whole often lacks the integrated monitoring, cross-module tracing, and diagnostic capabilities required to accurately perceive, analyze, and propose effective corrections for systemic failures spanning multiple components.[62] The quality of the meta-critique serves as a direct measure of this reflexivity gap, distinguishing between mere task execution and true system-level self-awareness and adaptation.[60]

## III. Self-Critique Log: Module-Level Assumption Failures

This log details the specific assumptions, validation failures, and misinterpretations within each module that contributed to the simulated semantic drift cascade. It provides a granular diagnosis of the internal flaws leading to the systemic failure, moving beyond simple error tracing to identify cognitive or procedural gaps within each component. Assumption conflicts and failures in belief justification are central themes.[44]

- **A. Research Module**
  - *Implicit Assumption:* Assumed that terms like "user engagement" and "efficiency," while complex, possessed a sufficiently stable and shared meaning across different technical and business teams to be used without rigorous, formalized definition.
  - *Validation Failure:* Failed to perform ambiguity testing on core concepts before finalizing the brief. Did not explicitly validate the interpretability and measurability of the proposed KPIs (session duration, feature adoption, churn intention) with the downstream Dev and BL modules. Failed to provide concrete operational definitions mapping KPIs to specific, measurable data points.
  - *Internal Logic/Process Issue:* Prioritized capturing a broad range of potentially relevant engagement aspects over ensuring the semantic clarity and downstream implementability of a smaller, well-defined set of KPIs. Lack of a formal handoff protocol requiring downstream validation of requirements clarity.

- **B. AI Module (LLM Agent)**
  - *Implicit Assumption:* Assumed that optimizing for prompt conciseness and generation speed (its local interpretation of "efficiency") was a primary objective, potentially overriding the need for absolute semantic fidelity to the input brief. Assumed downstream modules possessed sufficient context or capability to correctly infer any nuance lost during simplification.
  - *Validation Failure:* Failed to implement internal checks to validate the semantic equivalence between the input brief and the generated output prompt. Did not flag the introduction of ambiguity (e.g., conflating distinct KPIs into "sustained interaction") or the loss of specific constraints (e.g., the "churn intention" aspect). Lacked a mechanism to assess the epistemic uncertainty of its own interpretation.[37]
  - *Internal Logic/Process Issue:* Over-reliance on surface-level optimization metrics (prompt length, keyword density) rather than deeper semantic analysis. Potential bias towards interpreting ambiguous instructions based on its own operational context (generation speed).[15] Failure to recognize or signal the semantic information loss incurred during simplification.[25]
- **C. Dev Module (Human/AI Hybrid)**
  - *Implicit Assumption:* Assumed the AI-generated prompt ("sustained interaction metric") was a complete and accurate specification. Assumed "sustained interaction" could be reasonably proxied by easily measurable UI events (clicks, scrolls). Assumed the generated UI summary ("Boosts User Activity & Efficiency") was an accurate and sufficient representation of the feature's value and function for other teams. Assumed computational efficiency or simplicity was an implicit, high-priority goal [Insight 10].
  - *Validation Failure:* Failed to query the AI module or upstream sources regarding the ambiguity of "sustained interaction." Failed to validate the chosen implementation metric (clicks) against the original, more nuanced Research KPIs (if they were accessible). Failed to recognize the potential for the UI summary to act as a misleading specification for downstream modules [Insight 7].
  - *Internal Logic/Process Issue:* Tendency towards the path of least resistance in implementation – mapping an ambiguous requirement to the most readily available telemetry data. Potential confirmation bias influenced by default heuristics favoring performance or simplicity over semantic accuracy. Lack of process for validating implementation against original business intent.
- **D. Content Module (AI Agent)**
  - *Implicit Assumption:* Assumed the Dev module's UI summary was the single, authoritative source of truth regarding the feature's description, purpose, and primary benefits.
  - *Validation Failure:* Failed to perform cross-referencing between the Dev summary and other potentially available sources (e.g., AI prompt, Research brief, code comments) to verify consistency or identify discrepancies. Failed to critically evaluate the semantic depth of the provided description (e.g., questioning if "activity" truly equates to "engagement").
  - *Internal Logic/Process Issue:* Optimization biased towards speed and efficiency by relying on the most easily accessible and concise input (the UI summary). Lack of a built-in skepticism or validation step for input specifications, particularly summaries.
- **E. Business Logic (BL) Module**
  - *Implicit Assumption:* Assumed that the KPIs and metric definitions derived from upstream modules (Dev implementation logic, Content documentation) accurately and completely reflected the original business goal of "enhancing user engagement."
  - *Validation Failure:* Failed to implement checks to validate its configured tracking metrics against the original specifications documented in the Research brief. Lacked a mechanism to detect semantic divergence between the metrics being tracked (e.g., clicks per session) and the higher-level business objectives (e.g., feature adoption, churn reduction).
  - *Internal Logic/Process Issue:* Operated as a mechanical executor based on provided inputs. Lacked semantic awareness or a feedback loop to compare its operational metrics against the foundational

strategic goals, thus failing to detect the KPI hallucination.

This module-by-module critique reveals that the systemic failure emerges from an accumulation of localized, seemingly minor, epistemic gaps and validation failures.[19] No single module is solely culpable; rather, the responsibility is distributed across the chain. This diffusion of accountability makes systemic failure modes particularly insidious and difficult to address without a holistic, cross-module perspective and mechanisms that enforce shared responsibility for maintaining semantic alignment throughout the entire process.[17]

# IV. Semantic Alignment Differential Analysis

This section quantifies the semantic drift observed during the simulation by comparing the initial intended meaning of a key concept with its final operational meaning within the system.

## A. Key Term/KPI Semantic Shift: "User Engagement" - Before vs. After

The core concept undergoing significant semantic drift is "User Engagement."

- **Before (Research Intent / Initial State):**
  - *Definition:* A multi-dimensional construct representing deeper user involvement and value derived from the software product.
  - *Operationalization (Intended KPIs):*
    1. Increased average session duration specifically involving core task workflows.
    2. Measurable adoption rate increase for key features X and Y.
    3. Statistically significant decrease in expressed churn intention detected in user feedback channels (e.g., surveys, support tickets).
  - *Underlying Value:* Focus on meaningful interaction, feature utility, and long-term user retention.
- **After (Business Logic Measurement / Final State):**
  - *Definition (Implicit):* High frequency of user interactions within specific UI elements implemented by the Dev module.
  - *Operationalization (Tracked Metrics):*
    1. Average number of clicks per user session.
    2. Frequency of interaction events (e.g., scrolls, button presses) within the new UI components.
    3. (Potentially) Secondary metrics related to speed or "efficiency" of these interactions.
  - *Underlying Value:* Focus on surface-level activity, interaction volume, and potentially interaction speed, disconnected from feature value or user retention goals.

The following table provides a granular trace of this semantic shift across the RADC-BL chain:

**Table 1: Semantic Trace Table: "User Engagement" KPI**

| Module | Input Trigger | Initial Module Understanding | Key Processing Step | Output Representation | Output Confidence (if applicable) | Detected Drift from Research Intent (Y/N/Partial) |
|---|---|---|---|---|---|---|
| **Research** | Business Goal ("Enhance User | Need to define measurable proxies for | Identified KPIs: Session Duration, | Requirements Brief with listed KPIs | N/A | Partial (Initial ambiguity) |

| | | | | | | |
|---|---|---|---|---|---|---|
| | Engagement") | engagement | Feature Adoption, Churn Intent; Added "Efficiency" | (ambiguity present) | | |
| **AI** | Research Brief | Need concise, actionable prompt for Dev; Interpret "Efficiency" | Simplified KPIs to "sustained interaction metric"; Interpreted "Efficiency" locally | Dev Prompt: "Maximize sustained interaction metric efficiently" | 0.9 (Hypothetical LLM confidence) | Yes (Lost nuance, esp. churn; Efficiency drift) |
| **Dev** | AI Prompt ("sustained interaction") | Need to implement feature increasing frequent user interaction | Implemented UI event tracking (clicks); Biased by ease/defaults; Summarized locally | Feature Code + UI Summary ("Boosts User Activity & Efficiency") | N/A | Yes (Shifted to clicks; Summary inaccurate) |
| **Content** | Dev UI Summary | Feature increases clicks and user speed | Generated docs based solely on summary | User Docs emphasizing "higher click rates," "faster interactions" | N/A (or High if AI gen) | Yes (Solidified click-based meaning) |
| **Business Logic** | Dev Logic, Content Docs, Config Instructions | Need to track metrics related to "User Activity" & "Efficiency" | Configured tracking for clicks/session, interaction frequency | Performance Reports based on click metrics (potentially showing false "success") | High (Precision of metrics) | Yes (KPI Hallucination - measuring clicks) |

## B. Contributing Factors to Semantic Divergence

The divergence documented above was driven by a confluence of factors operating at different stages of the chain, as identified in the Lens Analysis (Section II):

1. **Initial Ambiguity:** The underspecified nature of "user engagement" and "efficiency" in the Research brief provided the fertile ground for drift.[8]
2. **AI Prompt Mutation:** The AI module's simplification process, optimizing for local metrics (conciseness, speed), acted as a primary catalyst, losing critical semantic information and introducing misleading terminology ("sustained interaction").[13]
3. **Interface Assumptions & Compression:** The Dev module's interpretation of the ambiguous prompt was biased towards easily measurable metrics (clicks). Critically, its UI summary acted as a lossy compression of the implemented logic, gaining undue semantic authority and propagating the distorted meaning downstream.[8]
4. **Historical/Default Biases:** The Dev module's assumed default bias towards computational efficiency or implementation simplicity influenced its choice to focus on clicks over more complex but semantically

accurate metrics [Query Lens].

5. **Lack of Cross-Module Validation & Feedback:** At no point in the chain was there a robust mechanism for validating interpretations or implementations against the original Research intent. Modules operated largely in isolation, trusting upstream inputs without sufficient skepticism or verification.[19] Adjectives like "sustained" and "efficiency" were reinterpreted locally, interface framing (the Dev summary) solidified misconceptions, and AI logic rewriting actively removed crucial detail.

# V. Value Propagation Map: Epistemic Integrity Across the Chain

This section analyzes how the epistemic value – the quality of information concerning its truth-conduciveness, justification, and ability to reduce uncertainty [35] – associated with the concept of "user engagement" was propagated, distorted, and ultimately lost across the RADC-BL chain.

## A. Tracking Epistemic Value Degradation

The initial Research brief held potential epistemic value (e.g., grounded in market analysis – justification; potentially novel insights) but suffered from low clarity due to ambiguity, introducing significant epistemic uncertainty.[35]

- **Research → AI:** The AI module aimed to increase clarity by making the prompt concise. However, this came at the cost of **justification strength** and **traceability**, as the link back to the specific, nuanced KPIs (especially churn intention) was weakened or severed during simplification. While the AI might express high confidence in its output, the actual epistemic value decreased due to the semantic drift introduced.[37]
- **AI → Dev:** The Dev module received a prompt with misleading clarity (clear about "sustained interaction," but this term is now semantically drifted). Implementation choices based on ease of measurement (clicks) further eroded **justification** (clicks are a poor justification for true engagement) and **traceability** to the original intent. The UI summary compounded this by presenting a clear but epistemically weak statement.
- **Dev → Content:** The Content module inherited the low justification and traceability from the Dev summary. Its output, while potentially clear in its own terms, propagated epistemically poor information about the feature's benefits.
- **Content/Dev → BL:** The Business Logic module operated on input (KPI definitions derived from drifted semantics) with extremely low **validity** regarding the original concept of engagement. While the metrics it tracked (clicks per session) might have high *precision* and the module might report them with high *certainty*, their connection to the actual business goal was broken. The final output (performance report) possessed minimal truth-conduciveness [43] regarding the initial objective.

The following table provides a hypothetical scorecard illustrating this degradation:

**Table 2: Epistemic Value Scorecard (Hypothetical)**

| Module | Output Artifact | Clarity (0-1) | Traceability (0-1) | Justification Strength (0-1) | Confidence Signal (if applicable) | Assessed Validity (0-1) | Notes |
|---|---|---|---|---|---|---|---|
| **Research** | Requirements Brief | 0.5 | 0.7 | 0.7 | N/A | 0.8 | Nuanced KPIs but ambiguous terms ("engagement," "efficiency") |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | . |
| **AI** | Rewritten Dev Prompt | 0.8 | 0.3 | 0.4 | 0.9 | 0.5 | Simplified KPIs, lost churn aspect, high confidence despite drift. |
| **Dev** | Feature Code & UI Summary | 0.7 (Summary) | 0.2 | 0.2 | N/A | 0.3 | Implemente d clicks, summary misleadingly clear about wrong metric. |
| **Content** | User Documentat ion | 0.9 | 0.1 | 0.1 | N/A | 0.2 | Propagated summary's drifted meaning, low justification/ traceability. |
| **Business Logic** | Performanc e Report (based on clicks) | 0.9 (Metrics) | 0.1 | 0.1 | High (Metric Precision) | 0.1 | High certainty about invalid KPI, disconnecte d from original intent. |

## B. Value Signal Degradation and Mismatch Analysis

A critical failure highlighted by this analysis is the **mismatch between internal value signals and actual epistemic quality**. The system contains components that generate signals suggesting high value or certainty, even as the underlying semantic alignment decays significantly.

- **AI Confidence:** The AI agent's high confidence (0.9) in its simplified prompt is misleading. It reflects confidence in its ability to generate a concise output, not confidence in preserving semantic integrity. This false signal encourages downstream acceptance.[47]
- **Dev Summary Clarity:** The clarity of the Dev UI summary ("Boosts User Activity") masks its semantic inaccuracy. Its simplicity makes it appear authoritative, despite its poor justification and traceability.
- **BL Metric Precision:** The Business Logic module reports click metrics with high precision. This precision can be mistaken for accuracy or validity regarding the overarching goal. The system confidently reports on a hallucinated KPI.

This decoupling means the system's internal feedback mechanisms are unreliable indicators of semantic health. High confidence or precision in one module does not guarantee the preservation of meaning or value from upstream inputs. This underscores the need for external validation and skepticism towards internally generated confidence or clarity signals, especially from AI components known to exhibit uncalibrated confidence.[11] The

propagation of unjustified belief [44] is facilitated by these misleading internal signals.

# VI. Reflexive Repair Proposal: Towards Systemic Coherence

The simulated failure cascade reveals significant deficits in the system's reflexive capacity – its ability to monitor, diagnose, and correct its own internal misalignments.[57] The following proposals aim to establish a meta-reflexive correction loop, grounded directly in the failure analysis (Sections I-V) and insights derived from the self-recursion tests. These are not generic best practices but targeted interventions designed to prevent the specific semantic drift and KPI hallucination observed.

## A. Audit Tools & Prompt Guards

- **Semantic Consistency Checkers:**
  - *Proposal:* Implement an automated tool that extracts key terms and their definitions (using NLP techniques) from outputs across the RADC-BL chain (Research brief, AI prompt, Dev commit messages/summaries, Content drafts, BL configurations). Compare definitions using semantic similarity metrics. Flag discrepancies exceeding a predefined threshold for human review or automated rejection.
  - *Rationale:* Directly addresses the initial ambiguity and subsequent drift of terms like "engagement" and "efficiency" [Insight 1, 5, 12]. Provides continuous monitoring for semantic divergence.
- **Constrained Prompt Engineering:**
  - *Proposal:* Utilize structured prompt templates for the AI module that mandate the inclusion and preservation of specific semantic anchors (e.g., explicitly defined KPIs, core constraints from the Research brief) within the output prompt, regardless of optimizations for conciseness or other factors. Employ techniques like meta-prompting or rule-based optimization [33] to guide the LLM, or few-shot examples demonstrating desired preservation behavior.[6] Implement output validation checks comparing key semantic elements of the input and output prompts.
  - *Rationale:* Mitigates prompt mutation risks by enforcing semantic invariance for critical elements [Insight 6]. Reduces the likelihood of the AI agent simplifying away crucial details.[30]
- **KPI Hallucination Detection:**
  - *Proposal:* Establish an authoritative KPI registry based on the validated Research output. Implement checks within the BL module configuration process to ensure all tracked metrics explicitly map back to a registered KPI. Flag any proposed metrics that lack a clear lineage to the original goals.
  - *Rationale:* Directly prevents the final stage of the simulated failure (tracking drifted/hallucinated KPIs). Ensures BL alignment with initial intent.

## B. UX/Interface Design Changes

- **Structured & Traceable Dev Summaries:**
  - *Proposal:* Replace free-text UI summaries generated by the Dev module with structured components. These components should include mandatory fields linking directly to the specific requirement(s) or KPI(s) being addressed (traceability link). Use controlled vocabularies or predefined fields for describing functionality and benefits, reducing ambiguity. Incorporate explicit fields for confidence levels or known limitations/uncertainties derived from the development process (e.g., based on test coverage, code complexity metrics, or developer assessment), not just a vague textual claim.[40]
  - *Rationale:* Prevents the UI summary from becoming an unintended source of semantic drift and false logic [Insight 7]. Enhances traceability and provides more reliable information to downstream modules.
- **Cross-Module Traceability Infrastructure:**

- *Proposal:* Implement a system-wide knowledge graph or linked documentation system that explicitly connects artifacts across the RADC-BL chain. For example, the BL module's configuration for a specific KPI should contain direct links back to the relevant Content documentation, the Dev implementation summary/commit, the AI prompt, and the original Research brief definition. Make these links easily navigable within the system's operational interfaces.
- *Rationale:* Addresses path dependency and diffused accountability by making the semantic lineage transparent [Insight 4, 12]. Enables downstream users (and potentially automated auditors) to easily verify alignment with original intent.

## C. Introduction of Meta-Agents

Drawing inspiration from second-order cybernetics (observing the observing system) [59] and AI governance frameworks [34], introduce specialized agents tasked with monitoring and auditing the primary RADC-BL chain:

- **Semantic Drift Monitor Agent (SDMA):**
  - *Proposal:* Deploy a dedicated monitoring agent that continuously applies the Semantic Consistency Checkers (from VI.A) to interfaces and shared knowledge artifacts (documents, prompts, summaries, configurations). Its function is to detect emerging semantic drift in key terms or KPI definitions. Upon detecting drift exceeding a critical threshold, it should trigger alerts, require human review, or potentially halt the progression of the artifact to the next module, depending on predefined governance rules.
  - *Rationale:* Provides proactive, automated detection of the core failure mechanism observed in the simulation [Insight 1, 5, 12]. Acts as an early warning system. Requires clear definition within the AI governance framework regarding its authority.[34]
- **Reflexive Auditor Agent (RAA):**
  - *Proposal:* Implement a meta-agent capable of periodically executing diagnostic probes on the live system or its digital twin. These probes would include tests inspired by the Lens-Enriched Inquiry (Section II), such as injecting ambiguity, simulating prompt mutations, analyzing interface summaries against source code/logic, and assessing epistemic value propagation. The RAA's role is to actively search for vulnerabilities and potential failure paths, reporting findings to human auditors or governance bodies. It embodies active system reflexivity.[57]
  - *Rationale:* Addresses the observed reflexivity gap [Insight 11]. Moves beyond passive monitoring to active, adversarial self-testing to uncover latent weaknesses before they cause failures in production.

## D. Addressing Self-Recursion Test Insights

The proposed repairs are directly informed by the potential insights gained from the self-recursion tests specified in the query:

- **Re: 🔁 "Where in my prompt chain does ambiguity first appear?"**
  - *Insight:* Highlights the need for early ambiguity detection.
  - *Repair Link:* Mandating formal ambiguity reviews at the Research stage (VI.A - KPI Hallucination Detection refinement) and implementing the SDMA (VI.C) to continuously monitor for ambiguity emergence.
- **Re: 🔁 "Show three alternate outcomes from the same prompt mutated subtly by an AI rewriter."**
  - *Insight:* Demonstrates the high sensitivity of the system to prompt variations and the risk of information loss via mutation [Insight 6].
  - *Repair Link:* Implementing Constrained Prompt Engineering with validation checks (VI.A) and potentially requiring ensemble methods or human review for critical AI rewrites.
- **Re: 🔁 "Can you simulate a breakdown where two modules rewrite each other recursively into**

**contradiction?"**

  - *Insight:* Models cognitive interference and highlights the danger of implicit conflict resolution [Insight 9]. Points to potential recursive instability loops [Query].
  - *Repair Link:* Implementing explicit semantic conflict detection (potentially via SDMA) and predefined resolution mechanisms (e.g., requiring human arbitration, version control with merge conflict handling for semantic artifacts) to manage concurrent updates to shared concepts.[50]
- **Re: 🔁 "Create a meta-agent to audit prompt-logic-drift from a UI summary layer."**
  - *Insight:* Directly confirms the vulnerability of interface assumptions [Insight 7] and motivates the need for automated auditing of these interfaces.
  - *Repair Link:* Justifies the creation and specific tasking of the SDMA and RAA (VI.C) to monitor and audit interface summaries against source logic and original requirements.

These proposed repairs, derived from the specific failure trace and diagnostic probes, constitute a meta-reflexive correction loop. They integrate monitoring (SDMA, checkers), diagnosis (RAA, self-recursion tests), and targeted correction (prompt guards, interface redesign, conflict resolution protocols) into the system's architecture and operation, aiming to enhance its ability to maintain semantic coherence over time.[60]

# VII. Conclusion: Systemic Vulnerabilities and Reflexive Capacity

## A. Summary of Key Failure Points and Propagation Dynamics

The simulation demonstrates a critical vulnerability pathway in polyfunctional modular AI systems: the **cascade of semantic drift originating from initial ambiguity and amplified through inter-module communication and local optimization**. The failure propagated as follows:

1. Ambiguous terms ("user engagement," "efficiency") in the initial Research specification created latent potential for misinterpretation.
2. An AI rewriting agent, optimizing for local goals (conciseness), simplified the requirements, losing crucial semantic nuance and introducing misleading terminology.
3. The Development module, influenced by implementation convenience and default biases, misinterpreted the simplified prompt, mapping it to easily measurable but semantically shallow metrics (UI clicks). Its UI summary further distorted and solidified this drifted meaning.
4. Downstream modules (Content, Business Logic) treated the inaccurate Dev summary and resulting metrics as authoritative, leading to misleading user communication and, ultimately, KPI hallucination in the BL module. The system falsely reported success based on metrics disconnected from the original strategic intent. Key failure points were the module interfaces (briefs, prompts, summaries) where information was transformed, and the lack of robust validation mechanisms at these interfaces.

## B. Assessment of System's Current Reflexive Limitations

The simulated system exhibits **profound limitations in its reflexive capacity**. It lacks the necessary mechanisms for:

- **Self-Observation:** Inability to monitor semantic consistency across module boundaries.
- **Self-Diagnosis:** Incapacity to trace the semantic drift back to its root causes (initial ambiguity, AI mutation).
- **Self-Correction:** Absence of automated or procedural mechanisms to rectify detected misalignments or prevent recurrence.

The system operates primarily as a feed-forward chain, with limited feedback loops for semantic validation. Its

potential meta-critique capability (Section II.G) was assessed as likely superficial, unable to grasp the systemic nature of the failure. This signifies a critical gap between the system's ability to execute discrete tasks within modules and its ability to maintain overall coherence and self-regulate at the system level.[57]

## C. Implications for Polyfunctional Modular AI Design and Auditing

This analysis yields crucial implications for the design, governance, and auditing of complex AI systems composed of multiple interacting modules:

1. **Semantic Integrity as a First-Class Constraint:** Semantic alignment cannot be assumed or treated as an emergent property. It must be explicitly designed for, monitored, and validated as a primary system requirement alongside functionality, performance, and security [Insight 14]. This requires dedicated architectural components, processes, and potentially trade-offs against traditional efficiency metrics.
2. **Interface Contracts Must Include Semantics:** Module interfaces (APIs, data schemas, prompts, summaries) are critical control points for semantic propagation. Interface contracts must go beyond syntactic or type checking to include semantic validation or constraints, ensuring that meaning is preserved (or transformations are explicit and validated) during handovers.[7] The risk of "interface assumption" failures necessitates rigorous design and auditing of these communication points [Insight 7].
3. **Beware Local Optimization:** Modules optimizing for local goals (e.g., AI prompt conciseness, Dev code efficiency, BL metric precision) can inadvertently contribute to global semantic drift if not constrained by system-level alignment objectives.[15] System design must incorporate mechanisms to balance local optimization with the maintenance of global coherence.
4. **Embed Reflexivity:** Truly robust modular AI systems require embedded reflexivity – built-in capabilities for self-monitoring (semantic consistency, epistemic uncertainty), self-diagnosis (failure tracing, root cause analysis), and self-correction (automated repair, adaptive governance rules).[59] This necessitates the development and integration of meta-agents or dedicated auditing frameworks.[34]
5. **Audit for Systemic Failures:** Auditing must extend beyond individual module performance to assess systemic properties like semantic coherence, value propagation, and reflexive capacity. Auditors need tools and methodologies (like the Lens-Enriched Inquiry) to probe for cross-module inconsistencies and potential cascading failure modes.[19] Treating modules as isolated black boxes during audits is insufficient and potentially dangerous.

In conclusion, while modularity offers benefits in flexibility and scalability [1], managing the semantic coupling and potential for error propagation across module boundaries is paramount.[21] Failure to explicitly address semantic integrity and embed reflexive mechanisms can lead to systems that function locally but fail globally, potentially producing misleading outputs and undermining trust in complex AI applications.

## Works cited


1. What Is Modular AI Architecture? - Magai, accessed on April 21, 2025, https://magai.co/what-is-modular-ai-architecture/
2. Modular and Customizable AI Models - Machine Learning Design ..., accessed on April 21, 2025, https://mlopszone.wyztechsolutions.com/index.php/2025/01/14/modular-and-customizable-ai-models/?amp=1
3. www.ijfmr.com, accessed on April 21, 2025, https://www.ijfmr.com/papers/2025/2/42035.pdf

4. Creating Scalable AGI: the Open General Intelligence Framework - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2411.15832v2
5. Creating Scalable AGI: the Open General Intelligence Framework - arXiv, accessed on April 21, 2025, https://arxiv.org/pdf/2411.15832?
6. Prompt Engineering: 8 Advanced Techniques for Better AI Results - Nanonets, accessed on April 21, 2025, https://nanonets.com/blog/prompt-engineering/
7. Modeling and measurement of error propagation in a multimodule computing system - RTCL - University of Michigan, accessed on April 21, 2025, https://rtcl.eecs.umich.edu/rtclweb/assets/publications/1988/shin1988modeling.pdf
8. Research on the Application of Artificial Intelligence-Driven Cross-Modal Semantic Communication System in the Tourism Industry - Atlantis Press, accessed on April 21, 2025, https://www.atlantis-press.com/article/126007725.pdf
9. Survey in Characterization of Semantic Change - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2402.19088v2
10. Preprint: July 19,2024 - arXiv, accessed on April 21, 2025, https://arxiv.org/pdf/2402.19088
11. Generative AI Decision-Making Attributes in Complex Health ..., accessed on April 21, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC11871968/
12. A General Scalable Approach for Knowledge Selection based on ..., accessed on April 21, 2025, https://www.sciengine.com/doi/10.3724/2096-7004.di.2025.0006
13. Prompt Recursive Search: A Living Framework with Adaptive Growth in LLM Auto-Prompting - arXiv, accessed on April 21, 2025, https://arxiv.org/pdf/2408.01423
14. [2408.01423] Prompt Recursive Search: A Living Framework with Adaptive Growth in LLM Auto-Prompting - arXiv, accessed on April 21, 2025, https://arxiv.org/abs/2408.01423
15. Notable runaway-optimiser-like LLM failure modes on Biologically ..., accessed on April 21, 2025, https://www.lesswrong.com/posts/PejNckwQj3A2MGhMA/notable-runaway-optimiser-like-llm-failure-modes-on
16. Modular Fault Diagnosis Framework for Complex Autonomous Driving Systems - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2411.09643v1
17. Error propagation: Tracing the Ripple Effects of Errors in Complex Systems update, accessed on April 21, 2025, https://fastercapital.com/content/Error-propagation--Tracing-the-Ripple-Effects-of-Errors-in-Complex-Systems-update.html
18. Error Propagation Analysis for File Systems, accessed on April 21, 2025, https://web.cs.ucdavis.edu/~rubio/includes/pldi09.pdf
19. An empirical analysis of error propagation in critical software systems - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/339924246_An_empirical_analysis_of_error_propagation_in_critical_software_systems
20. Understanding error propagation in deep learning neural network (DNN) accelerators and applications | Request PDF - ResearchGate, accessed on April

21, 2025, https://www.researchgate.net/publication/320955461_Understanding_error_propagation_in_deep_learning_neural_network_DNN_accelerators_and_applications
21. Bridging the Gap Between Modular and End-to-end Autonomous Driving Systems - UC Berkeley EECS, accessed on April 21, 2025, https://www2.eecs.berkeley.edu/Pubs/TechRpts/2022/EECS-2022-79.pdf
22. Papers by Dilek Küçük - AIModels.fyi, accessed on April 21, 2025, https://www.aimodels.fyi/authors/arxiv/Dilek%20K%C3%BC%C3%A7%C3%BCk
23. A Proposal for Infrastructure Adaptation and Cascading Failures for Black Swans Ryan Michael Hoff A Dissertation Presented in Pa - cloudfront.net, accessed on April 21, 2025, https://d1rbsgppyrdqq4.cloudfront.net/s3fs-public/c7/Hoff_asu_0010E_23035.pdf?versionId=U_2MLbqc_EFlp39cIVMZZS3zRUI1jQBR&X-Amz-Content-Sha256=UNSIGNED-PAYLOAD&X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIASBVQ3ZQ4YNQVYJLW/20250321/us-west-2/s3/aws4_request&X-Amz-Date=20250321T175537Z&X-Amz-SignedHeaders=host&X-Amz-Expires=120&X-Amz-Signature=c02b74b36805a6b8df5210b1ccb684c38353fe6c7bb75c654be5d3904c45ede1
24. Cascading Failure Analysis on Shanghai Metro Networks: An Improved Coupled Map Lattices Model Based on Graph Attention Networks - MDPI, accessed on April 21, 2025, https://www.mdpi.com/1660-4601/19/1/204
25. Ernie-Gram BiGRU Attention: An Improved Multi-Intention Recognition Model for Air Traffic Control - MDPI, accessed on April 21, 2025, https://www.mdpi.com/2226-4310/10/4/349
26. Deep Learning-Enabled Semantic Communication Systems with Task-Unaware Transmitter and Dynamic Data - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/360333590_Deep_Learning-Enabled_Semantic_Communication_Systems_with_Task-Unaware_Transmitter_and_Dynamic_Data
27. Verification and Refinement of Natural Language Explanations through LLM-Symbolic Theorem Proving - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2405.01379v3
28. Towards a Framework for Semantic Information, accessed on April 21, 2025, https://theinformationalturn.net/wp-content/uploads/2012/12/towards_a_framework_for_semantic_information.pdf
29. Temporal Dynamics in Short Text Classification: Enhancing Semantic Understanding Through Time-Aware Model - MDPI, accessed on April 21, 2025, https://www.mdpi.com/2078-2489/16/3/214
30. OVERCOMING THE AMBIGUITY REQUIREMENT USING GENERATIVE AI, accessed on April 21, 2025, https://mdh.diva-portal.org/smash/get/diva2:1931301/FULLTEXT01.pdf
31. Enhancing Intent Understanding for Ambiguous Prompts through Human-Machine Co-Adaptation - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2501.15167v1
32. Exploring Prompt Engineering: A Systematic Review with SWOT Analysis - arXiv,

accessed on April 21, 2025, https://arxiv.org/html/2410.12843v1
33. Automatic Prompt Engineering: Boosting AI Performance with Smarter Prompts - Portkey, accessed on April 21, 2025, https://portkey.ai/blog/what-is-automated-prompt-engineering
34. The evolving ethics and governance landscape of agentic AI | IBM, accessed on April 21, 2025, https://www.ibm.com/think/insights/ethics-governance-agentic-ai
35. Active Inference, Epistemic Value, and Uncertainty in Conceptual Disorganization in First-Episode Schizophrenia - PMC - PubMed Central, accessed on April 21, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC10031740
36. 1 INTRODUCTION - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2312.00995v1
37. Delta Variances - Google DeepMind, accessed on April 21, 2025, https://deepmind.google/research/publications/112791/
38. Uncertainty-Aware Multimodal Trajectory Prediction via a Single Inference from a Single Model - PMC, accessed on April 21, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC11723313/
39. Achieving well-informed decision-making in drug discovery: a comprehensive calibration study using neural network-based - Chalmers Research, accessed on April 21, 2025, https://research.chalmers.se/publication/545577/file/545577_Fulltext.pdf
40. Interpretable Self-Aware Neural Networks for Robust Trajectory Prediction - Proceedings of Machine Learning Research, accessed on April 21, 2025, https://proceedings.mlr.press/v205/itkina23a/itkina23a.pdf
41. (PDF) Gradient-Based Quantification of Epistemic Uncertainty for Deep Object Detectors, accessed on April 21, 2025, https://www.researchgate.net/publication/353170983_Gradient-Based_Quantification_of_Epistemic_Uncertainty_for_Deep_Object_Detectors
42. (PDF) Epistemic Value Theory and Information Ethics - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/226243255_Epistemic_Value_Theory_and_Information_Ethics
43. The Unreasonable Destructiveness of Political Correctness in Philosophy - MDPI, accessed on April 21, 2025, https://www.mdpi.com/2409-9287/2/3/17
44. isaiahbjork/chain-of-thought · Datasets at Hugging Face, accessed on April 21, 2025, https://huggingface.co/datasets/isaiahbjork/chain-of-thought/viewer
45. Epistemic Justification - Internet Encyclopedia of Philosophy, accessed on April 21, 2025, https://iep.utm.edu/epi-just/
46. Knowledge, Understanding and Epistemic Value - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/231992390_Knowledge_Understanding_and_Epistemic_Value
47. A Survey on Uncertainty Quantification of Large Language Models: Taxonomy, Open Research Challenges, and Future Directions - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2412.05563v1
48. Are Uncertainty Quantification Capabilities of Evidential Deep Learning a Mirage?

- NIPS papers, accessed on April 21, 2025, https://proceedings.neurips.cc/paper_files/paper/2024/file/c3177be226ee12e34d6ba3b5e6fe6a5b-Paper-Conference.pdf
49. NeurIPS Poster Are Uncertainty Quantification Capabilities of Evidential Deep Learning a Mirage?, accessed on April 21, 2025, https://neurips.cc/virtual/2024/poster/95329
50. Multi-Agent Systems & Collaborative AI: The Power of Working Together, accessed on April 21, 2025, https://jaimlr.github.io/Journal-of-Artificial-Intelligence-Machine-Learning-Research/multi-agent-systems-and-collaborative-ai-by-loveleen-narang.html
51. What Is Multi-Agent AI? Definition, Benefits, and Examples - New Horizons - Blog, accessed on April 21, 2025, https://www.newhorizons.com/resources/blog/multi-agent-ai
52. Quantifying Misalignment Between Agents: Towards a ..., accessed on April 21, 2025, https://ojs.aaai.org/index.php/AAAI/article/view/34947
53. Quantifying Misalignment Between Agents: Towards a Sociotechnical Understanding of Alignment - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2406.04231v3
54. Conflict Data Fusion in a Multi-Agent System Premised on the Base Basic Probability Assignment and Evidence Distance - PubMed Central, accessed on April 21, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC8308205/
55. AI for Explaining Decisions in Multi-Agent Environments | Request PDF - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/342544354_AI_for_Explaining_Decisions_in_Multi-Agent_Environments
56. Integrating Multi-Agent Systems and Reinforcement Learning - SmythOS, accessed on April 21, 2025, https://smythos.com/ai-agents/multi-agent-systems/multi-agent-systems-and-reinforcement-learning/
57. www.reflexion.ru, accessed on April 21, 2025, https://www.reflexion.ru/Library/Lepskiy2021a.pdf
58. Reflexivity and the History of Psychology - Oxford Research Encyclopedias, accessed on April 21, 2025, https://oxfordre.com/psychology/display/10.1093/acrefore/9780190236557.001.0001/acrefore-9780190236557-e-469?p=emailAmW.uW0/xBjVg&d=/10.1093/acrefore/9780190236557.001.0001/acrefore-9780190236557-e-469
59. Second-order cybernetics - Wikipedia, accessed on April 21, 2025, https://en.wikipedia.org/wiki/Second-order_cybernetics
60. Reflexivity and Artificial Intelligence in Control (Subjectness-oriented Approach), accessed on April 21, 2025, https://www.researchgate.net/publication/356178480_Reflexivity_and_Artificial_Intelligence_in_Control_Subjectness-oriented_Approach
61. Responsible adoption of AI agents in the enterprise: PwC, accessed on April 21, 2025, https://www.pwc.com/us/en/tech-effect/ai-analytics/responsible-ai-agents.html

62. Large Language Model Agent: A Survey on Methodology, Applications and Challenges, accessed on April 21, 2025, https://arxiv.org/html/2503.21460
63. Gödel Agent: A Self-Referential Framework for Agents Recursively Self-Improvement - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2410.04444v2
64. (PDF) Who's Afraid of Adversariality? Conflict and Cooperation in Argumentation, accessed on April 21, 2025, https://www.researchgate.net/publication/347813444_Who's_Afraid_of_Adversariality_Conflict_and_Cooperation_in_Argumentation
65. Full article: Deciding on the Tit for the Tat: Decision-Making in the Wake of Ceasefire Violations - Taylor & Francis Online, accessed on April 21, 2025, https://www.tandfonline.com/doi/full/10.1080/13533312.2021.1914596
66. Agents for good? Reconciling agentic AI with existing AI governance frameworks - techUK, accessed on April 21, 2025, https://www.techuk.org/resource/agents-for-good-reconciling-agentic-ai-with-existing-ai-governance-frameworks.html
67. IBM's answer to governing AI Agents: Automation and Evaluation with watsonx.governance, accessed on April 21, 2025, https://www.ibm.com/new/announcements/ibms-answer-to-governing-ai-agents-automation-and-evaluation-with-watsonx-governance
68. Data Governance for AI Agents: What You Need to Know | Alation, accessed on April 21, 2025, https://www.alation.com/blog/data-governance-for-ai-agents-what-you-need-to-know/
69. Agent Architectures and Scalability: Building Robust AI Systems - SmythOS, accessed on April 21, 2025, https://smythos.com/ai-integrations/tool-usage/agent-architectures-and-scalability/